
Template Answers for Quiz 1 for MT 3154 Graph Theory

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1. (a) Prove that a connected graph G is Eulerian if and only if the degree of every vertex of G is even. 4

[1] Suppose first that G is Eulerian. In an Euler tour, whenever we enter a vertex along an edge, we must leave it along another edge. Since the tour is closed, the incident edges at every vertex can therefore be paired as an entering edge and a leaving edge. Hence every vertex has even degree.

[0.5] Conversely, suppose that G is connected and every vertex of G has even degree. We prove that G has an Euler tour by induction on $|E(G)|$. The case $|E(G)| = 0$ is immediate.

[0.5] Starting with any edge and continuing along unused edges produces a non-trivial closed trail: a maximal such trail cannot get stuck at a vertex different from its starting vertex, since at such a vertex an odd number of incident edges would have been used, while its total degree is even.

[1] Let W be a closed trail of maximum length with no repeated edge, and let $F = E(W)$. If $F \neq E(G)$, consider $G' = G - F$. At every vertex, W uses an even number of incident edges, so every vertex of G' again has even degree. Since G is connected and G' has an edge, some component K of G' containing an edge meets W .

[1] By induction, K has an Euler tour. Starting this tour at a vertex of $K \cap W$ and inserting it into W gives a longer closed trail with no repeated edge, contradicting the maximality of W .

Thus $F = E(G)$, so W is an Euler tour. Hence G is Eulerian.

- (b) Prove that a graph G is a tree if and only if any two vertices of G are joined by a unique path in G . 3

Recall that a tree is a connected acyclic graph.

[1.5] Suppose G is a tree. Since G is connected, any two vertices x, y are joined by at least one x - y path. If there were two distinct x - y paths, their union would contain a cycle, contradicting that G is acyclic. Hence the x - y path is unique.

[1.5] Conversely, suppose that any two vertices of G are joined by a unique path. In particular, G is connected.

If G contained a cycle C , then any two distinct vertices $x, y \in V(C)$ would be joined by two distinct paths along the two directions around C , contradicting uniqueness.

Thus G is connected and acyclic, and hence G is a tree.

2. (a) Define perfect matching.

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For a graph $G = (V, E)$, a perfect matching is a subset M of edges in E such that every vertex in V is incident to exactly one edge in M .

- (b) Prove that, for every integer $k \geq 1$, every k -regular bipartite graph contains a perfect matching.

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Let G be a k -regular bipartite graph with bipartition (A, B) , where $k \geq 1$.

[0.5] Counting the edges from the two sides gives $k|A| = |E(G)| = k|B|$. Since $k \geq 1$, we have $|A| = |B|$.

[0.5] Precise statements of Hall's theorem.

[1] We verify Hall's condition. Let $S \subseteq A$. Exactly $k|S|$ edges have one endpoint in S , and all their other endpoints lie in $N(S)$.

Since every vertex of $N(S)$ has degree k , at most $k|N(S)|$ edges are incident with $N(S)$. Therefore $k|S| \leq k|N(S)|$, and hence $|S| \leq |N(S)|$.

Thus Hall's condition holds for every $S \subseteq A$.

[1] By Hall's theorem, G has a matching saturating A . Since $|A| = |B|$, this matching also saturates B . Hence it is a perfect matching.

3. Let $k \geq 1$ be an integer. Let G be a graph containing a cycle C , and suppose that G contains a path of length at least k whose endvertices lie on C . Prove that G contains a cycle of length at least $\sqrt{k}$.

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Let P be a path of length at least k whose endvertices lie on C .

[1] Suppose, for a contradiction, that every cycle in G has length strictly less than $\sqrt{k}$. In particular, $|C| < \sqrt{k}$.

List the vertices of $P \cap C$, in their order along P , as $x_0, x_1, \dots, x_r$.

Since these vertices are distinct vertices of C , $r + 1 \leq |C|$, and hence $r < \sqrt{k}$.

[1] For $i = 1, \dots, r$, let $Q_i = x_{i-1}Px_i$.

By the choice of consecutive vertices x_{i-1}, x_i , the interior of Q_i contains no vertex of C .

Thus $P = Q_1 \cup Q_2 \cup \dots \cup Q_r$ as a concatenation of these paths.

[1] We claim that every Q_i has length less than $\sqrt{k}$.

If Q_i is an edge of C , this is immediate since $1 < |C| < \sqrt{k}$. Otherwise, Q_i , together with either of the two $x_{i-1}-x_i$ paths in C , contains a cycle. By our assumption this cycle has length less than $\sqrt{k}$, and therefore $|E(Q_i)| < \sqrt{k}$.

[1] Consequently,

$$|E(P)| = \sum_{i=1}^r |E(Q_i)| < r\sqrt{k} < \sqrt{k}\sqrt{k} = k.$$

This contradicts the assumption that P has length at least k .

Hence G contains a cycle of length at least $\sqrt{k}$.

4. Prove that a graph G is 2-connected if and only if there exist subgraphs $H_0 \subseteq H_1 \subseteq \dots \subseteq H_m = G$ such that H_0 is a cycle and, for every $i \geq 1$, $H_i = H_{i-1} \cup P_i$, where P_i is an H_{i-1} -path.

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[0.5] First suppose that G is obtained from a cycle by successively adding H -paths. A cycle is 2-connected.

[1.5] It remains to observe that adding an H -path preserves 2-connectedness.

Let H be 2-connected and let P be an H -path with endvertices $x, y \in H$. Put $H' = H \cup P$. For any vertex $v \in V(H')$:

- if $v \in V(H)$, then $H - v$ is connected, and $P - v$ is still attached to $H - v$ through at least one of x, y ;
- if v is an inner vertex of P , then H remains connected, and the two pieces of $P - v$ are attached to H at x and y .

Hence $H' - v$ is connected for every v , so H' is 2-connected. By induction, G is 2-connected.

[0.5] Conversely, suppose that G is 2-connected. Then G contains a cycle. Choose a maximal subgraph $H \subseteq G$ that can be constructed from a cycle by successively adding H -paths.

[0.5] We first observe that H is an induced subgraph of G . Indeed, if

$$xy \in E(G) \setminus E(H) \quad \text{with} \quad x, y \in V(H),$$

then the edge xy itself is an H -path. Adding it to H would give a larger constructible subgraph, contradicting the maximality of H .

[1.5] Suppose that $H \neq G$. Since G is connected, there is an edge $vw \in E(G)$ with $v \notin V(H)$ and $w \in V(H)$.

Since G is 2-connected, $G - w$ is connected. Therefore $G - w$ contains a path from v to a vertex of $H - w$. Choose such a path P that first meets H at its last vertex.

Then wvP is an H -path: its two endvertices lie in H , while all its inner vertices lie outside H .

[0.5] Hence $H \cup wvP$ is a larger constructible subgraph of G , contradicting the maximality of H .

Therefore $H = G$, and G can be constructed from a cycle by successively adding H -paths.